Mahnum Zahid

AI Enthusiast | Data Science
+92-3044030190 | mahnumzahid161@gmail.com | Lahore, Pakistan

OBJECTIVE

Aspiring to secure a role that allows me to actively participate in real projects, apply my skills across the full development process, collaborate with diverse teams, and contribute to delivering effective, user-centered solutions while gaining hands-on experience and continuous learning. As a Data Science graduate and Associate AI Engineer, I want to apply my knowledge across the complete lifecycle of a data or AI solution, from data collection and preprocessing to model building, evaluation, deployment and monitoring, and to keep growing in Machine Learning, Natural Language Processing and Generative AI.

PROFESSIONAL SUMMARY

Associate AI Engineer at Salestech Data & AI Pakistan with a B.S. in Data Science from PUCIT, University of the Punjab. Experienced in building, evaluating and deploying Machine Learning and Generative AI (LLM-based) components using Python. Hands-on with data pipelines, Scikit-learn modelling, NLP techniques, prompt engineering and LLM APIs, and comfortable working in agile teams with sprint planning and code reviews. Also experienced in building full-stack, AI-driven applications and interactive dashboards through academic and personal projects.

Core competencies

- **Machine Learning:** model training, evaluation, hyperparameter tuning and deployment.
- **Natural Language Processing:** text preprocessing, tokenization and sentiment analysis.
- **Generative AI:** LLM APIs, prompt engineering and LLM-based application components.
- **Data engineering and analysis:** end-to-end pipelines, data cleaning, analysis and visualization.
- **Application development:** React front-ends, backend and database integration, Streamlit apps.
- **Collaboration:** agile delivery, sprint planning, code reviews and cross-functional teamwork.

PROFESSIONAL EXPERIENCE

Associate AI Engineer — Salestech Data & AI Pakistan *Present | Lahore, Pakistan*

Continuing at Salestech after completing the internship, working as a full-time member of the AI team on real client and product projects.

Solution design and development

- Designing and developing AI-powered solutions and data-driven features for real client and product projects, translating business and product requirements into working technical solutions.
- Working through the full lifecycle of each feature: understanding the problem, preparing and exploring data, building and testing the solution, and delivering it for use.
- Focusing on user-centered solutions so that the AI components are practical, reliable and useful to the clients and end users they serve.

Machine Learning and Generative AI

- Building, evaluating, and deploying Machine Learning components using Python, including data preprocessing, model training and performance evaluation.

- Developing and integrating Generative AI (LLM-based) components, applying prompt engineering and large language model APIs to solve real use cases.
- Evaluating model and LLM outputs for quality and consistency before components are deployed into products.

Teamwork and delivery

- Collaborating with cross-functional teams in an agile environment, including engineers, data professionals and product stakeholders.
- Contributing to code reviews, giving and receiving feedback to maintain code quality and share best practices.
- Taking part in sprint planning and delivery, helping the team meet milestones and release working features.

Tools and technologies: Python, Scikit-learn, Pandas, NumPy, LLM APIs, prompt engineering, agile workflow with code reviews.

Data Science Intern — Salestech Data & AI Pakistan *February 2026 – April 2026 | Lahore, Pakistan*

Three-month internship on the Data & AI team, working on practical data science deliverables across data engineering, machine learning, NLP and Generative AI.

Data processing and pipelines

- Applied Python for data processing, cleaning, transformation, scripting, and building end-to-end data pipelines.
- Built pipelines that take raw data through preprocessing and feature preparation to model-ready outputs.

Machine Learning

- Worked on ML tasks including model training, evaluation, and hyperparameter tuning using Scikit-learn.
- Compared model performance using appropriate evaluation metrics and tuned parameters to improve results.

Natural Language Processing

- Implemented NLP techniques (text preprocessing, tokenization, sentiment analysis) on real datasets.
- Prepared and cleaned unstructured text data so that it could be used reliably for analysis and modelling.

Generative AI

- Contributed to Gen AI experiments using large language model APIs and prompt engineering.
- Iterated on prompts and tested model responses to understand how LLMs behave on real tasks.

Agile practices

- Participated in sprint planning, code reviews, and delivery of data science deliverables as part of the team.

Tools and technologies: Python, Pandas, NumPy, Scikit-learn, NLP libraries, LLM APIs.

EDUCATION

B.S. Data Science – PUCIT, University of the Punjab, Lahore *2022 – 2026*

Four-year undergraduate degree at the Punjab University College of Information Technology (PUCIT), combining computer science, mathematics, statistics and applied data science.

Core areas of study

- **Programming and software:** Programming Fundamentals, Object-Oriented Programming, Data Structures and Algorithms, Software Engineering, Web Development.
- **Data and databases:** Database Systems (SQL), Data Warehousing, Data Mining, Big Data concepts.
- **Mathematics and statistics:** Calculus, Linear Algebra, Discrete Mathematics, Probability and Statistics.
- **Machine Learning and AI:** Machine Learning, Artificial Intelligence, Natural Language Processing, Generative AI, Deep Learning fundamentals.
- **Analytics and visualization:** Data Analysis, Data Visualization, Exploratory Data Analysis, dashboarding with Power BI.

Year-wise coursework

- **Year 1 (Foundations):** Programming Fundamentals, Calculus, Linear Algebra, Discrete Mathematics, Functional English.
- **Year 2 (Core computing):** Object-Oriented Programming, Data Structures and Algorithms, Database Systems, Probability and Statistics.
- **Year 3 (Data science core):** Machine Learning, Data Mining, Artificial Intelligence, Data Visualization, Software Engineering, Web Development.
- **Year 4 (Advanced and capstone):** Natural Language Processing, Deep Learning, Generative AI, Big Data, Final Year Project.

Key learning outcomes

- Ability to take a problem from raw data to a trained, evaluated and deployed model.
- Strong foundation in the mathematics and statistics behind machine learning algorithms.
- Practical experience building full applications, from user interface to database to AI features.
- Experience presenting data insights through charts and interactive dashboards.

Capstone: Final Year Project, *Notescape — AI-Powered Personal Learning Workspace* (see Personal Projects).

Skills developed: Python, SQL, Java, C/C++, HTML/CSS, Pandas, NumPy, Scikit-learn, Matplotlib, Streamlit, MySQL.

F.Sc. (Pre-Medical) – Unique College for Girls, Lahore *2018 – 2022*

Intermediate in the Pre-Medical group, building a strong foundation in the natural sciences and in analytical and scientific reasoning, which later supported the move into data science.

Subjects: Biology, Chemistry, Physics, English, Urdu, Islamiyat, Pakistan Studies.

Matriculation – Divisional Public School, Model Town, Lahore *2006 – 2018*

Secondary school education in the Science group, completed at Divisional Public School, Model Town, Lahore, giving a solid base in mathematics and the sciences.

Subjects: Physics, Chemistry, Biology, Mathematics, English, Urdu, Islamiyat, Pakistan Studies.

PERSONAL PROJECTS

Final Year Project | Notescape — AI-Powered Personal Learning Workspace *Completed*

Overview: Notescape is an AI-powered personal learning workspace designed to help students organize their study material, follow structured study workflows and track their learning progress in one place. The project combines a modern front-end, a backend with a database, and AI-driven features. Completed as the Final Year Project of the B.S. Data Science degree at PUCIT.

My role: Front-end development and front-end to backend integration.

Front-end development

- Developed the complete front-end user interface in React, building reusable components and screens.
- Implemented core study workflows and screens, including creating, organizing and reviewing learning content within the workspace.
- Focused on a clean, intuitive and user-centered experience so that students can navigate the platform easily.

Dashboards and analytics

- Created interactive dashboards and analytics views to visualize user progress and learning patterns.
- Presented study activity and progress in a clear visual form so that users can understand their habits and improve over time.

Integration

- Integrated front-end with backend and database to manage content, user data, and AI-driven features.
- Connected the UI to backend services so that user content, accounts and AI functionality work together as one application.

Key outcomes

- Delivered a working end-to-end application combining a user interface, backend, database and AI features.
- Gained hands-on experience in component-based front-end design, state management and API integration.
- Learned to present analytics in a way that is useful for non-technical users.

Technologies: React, HTML/CSS, backend APIs, database, AI/LLM-driven features, data visualization.

Uber Ride Prediction App

- Built a fare prediction model with Python, Pandas, and Scikit-learn; deployed with Streamlit.
- Performed data cleaning, feature preparation, model training and evaluation as part of the machine learning workflow.
- Deployed the model as an interactive web application so that users can enter trip details and receive fare predictions.

Technologies: Python, Pandas, Scikit-learn, Streamlit.

Covid-19 Data Analysis Dashboard

- Analyzed and visualized Covid-19 data using Pandas and Matplotlib; built interactive dashboards in Power BI.

- Cleaned and aggregated the data and performed trend analysis to identify key patterns.
- Presented insights through interactive Power BI dashboards that are easy to explore.

Technologies: Python, Pandas, Matplotlib, Power BI.

Library Management System

- Console-based application for book issuance, returns, and record management using OOP in C++.
- Used classes and objects to model books, members and transactions in a structured, maintainable way.

Technologies: C++, Object-Oriented Programming.

TECHNICAL SKILLS

Languages: Python, SQL, Java, C/C++, HTML/CSS

Data & ML: Machine Learning, Data Analysis, Data Visualization, NLP, Generative AI

Libraries/Tools: Pandas, NumPy, Scikit-learn, Matplotlib, Streamlit, MySQL

Additional tools: React, Power BI, LLM APIs, prompt engineering

Professional practices: Agile and sprint planning, code reviews, cross-functional collaboration, end-to-end data pipelines